Regression Diagnostics

POLS 602

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Dr. Mike Burnham
Texas A&M Political Science

Non-normal residuals

Why normal residuals matter

- Hypothesis tests (**not** OLS) rely on assuming the sampling distribution is roughly normal
- If sample size is small, CLT does not apply and we cannot assume normality
- We can estimate the shape of our sampling distribution by looking at the distribution of the errors.

$$\hat{\beta} = (X'X)^{-1}X'Y$$

$$\hat{\beta} = (X'X)^{-1}X'(X\beta + \epsilon)$$

$$\hat{\beta} = \beta + (X'X)^{-1}X'\epsilon$$

- β and X are fixed while ϵ is random. Thus the sampling behavior of $\hat{\beta}$ is determined by ϵ
- If model residuals are non-normal, $\hat{\beta}$ is unbiased but hypothesis tests about $\hat{\beta}$ may be unreliable

Causes

- True error distribution is non-normal
- Heteroskedasticity
- Outliers
- Small sample size
- Using a linear model for discrete outcomes

Testing your residuals

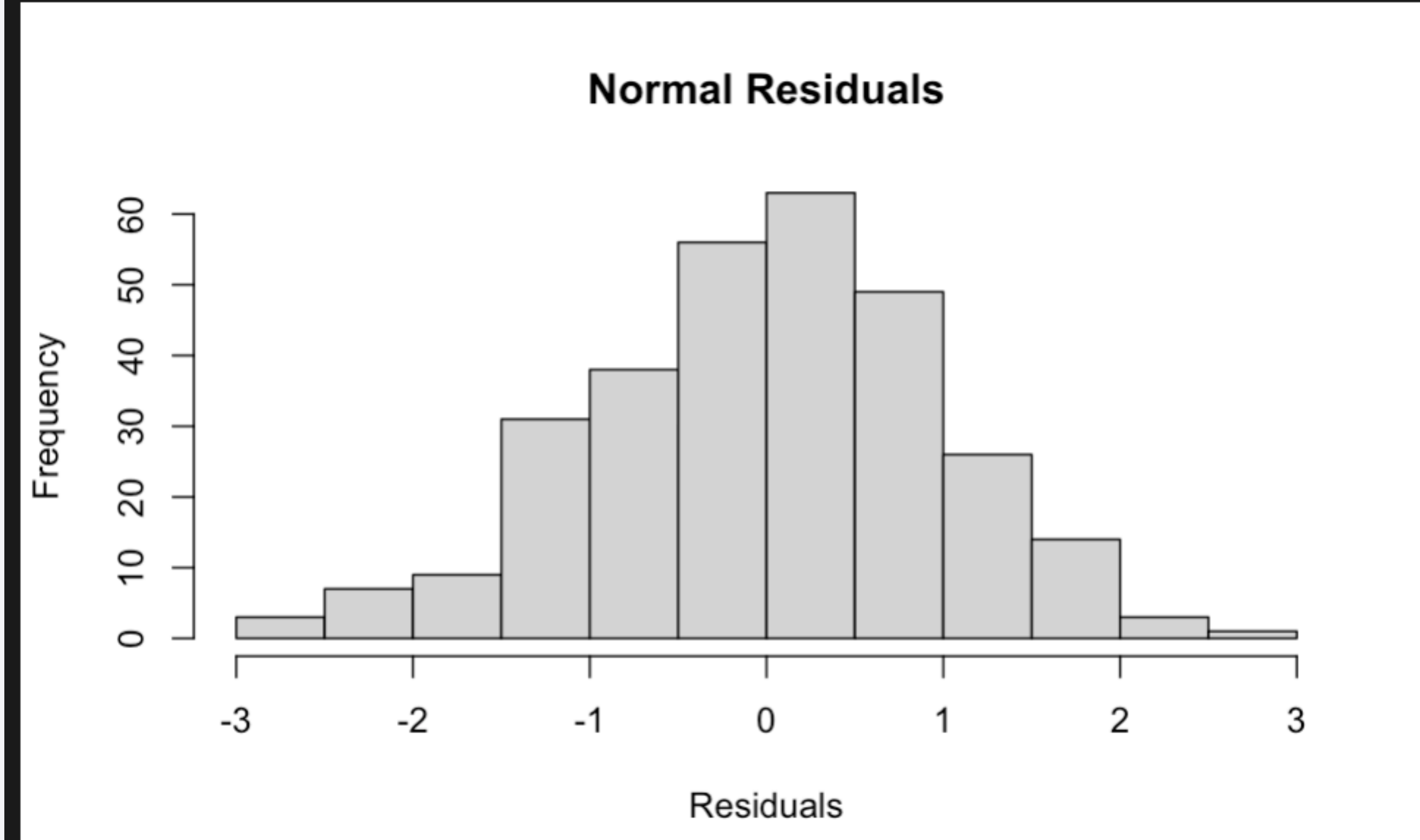
````
```{r}
# Simulate independent variable
n <- 300
x <- rnorm(n)

# Linear DGP with normal errors
errors_normal <- rnorm(n, mean = 0, sd = 1)
y_normal <- 2 + 3*x + errors_normal

# Fit model
model_normal <- lm(y_normal ~ x)
````
````

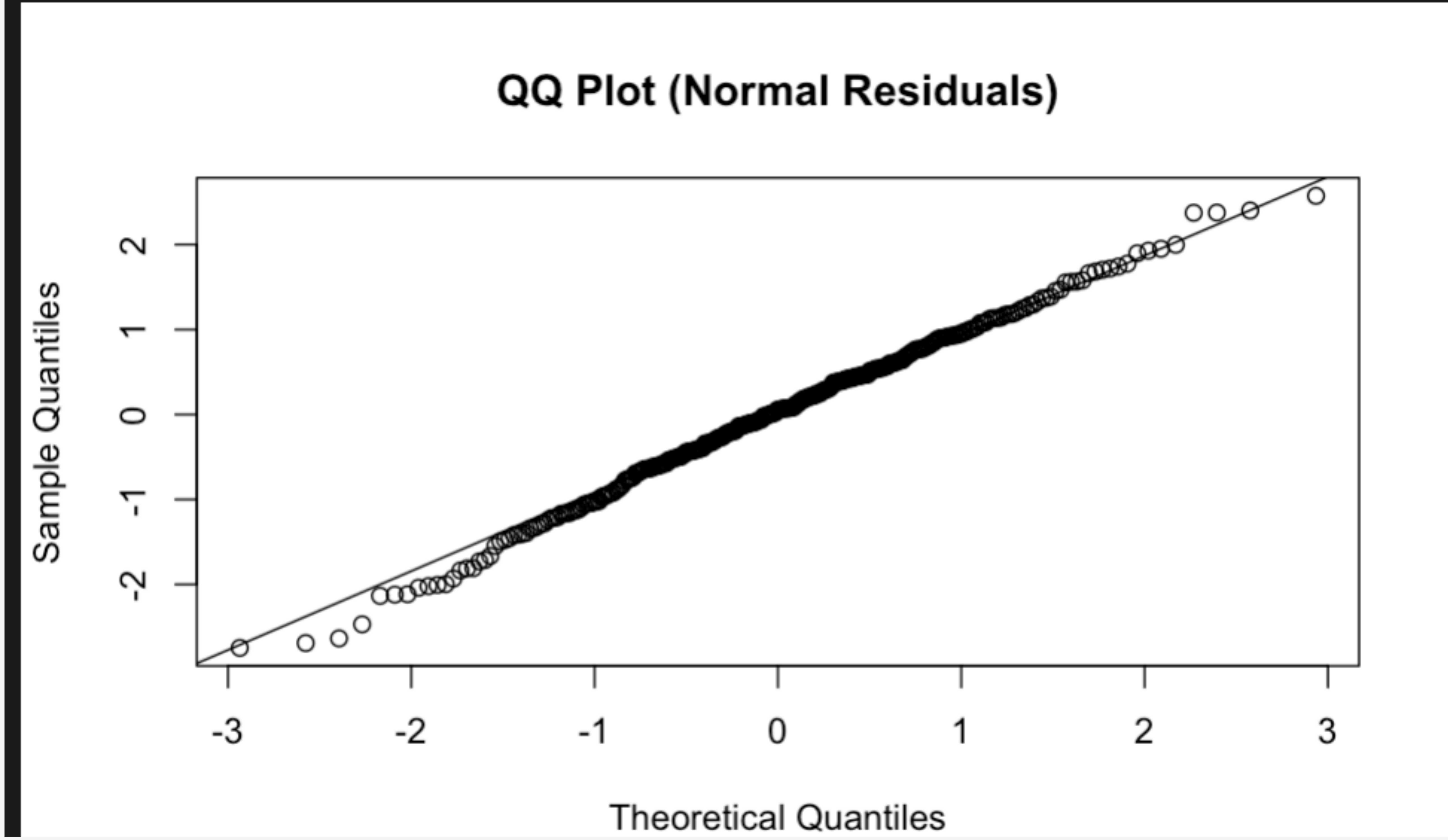
Testing your residuals

```
```{r}
# Histogram of the residuals
hist(resid(model_normal), main = "Normal Residuals", xlab = "Residuals")
```
```



Testing your residuals

```{r}
# QQ plot
qqnorm(resid(model_normal), main = "QQ Plot (Normal Residuals)")
qqline(resid(model_normal))
```



Testing your residuals

```{r}
# Shapiro-Wilk test
shapiro_normal <- shapiro.test(resid(model_normal))
shapiro_normal
```

```
	Shapiro-Wilk normality test

data:  resid(model_normal)
W = 0.99567, p-value = 0.576
```

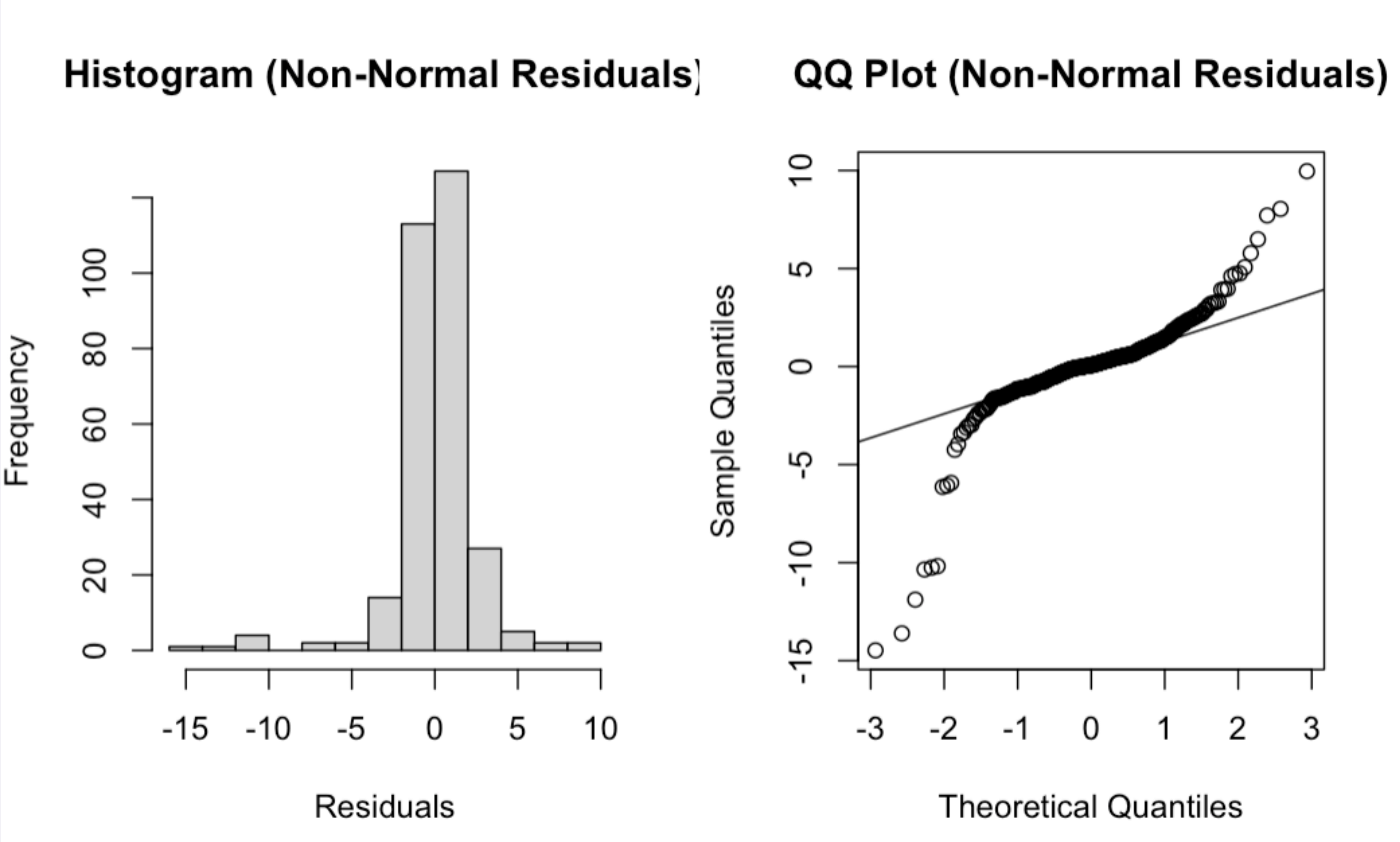
Testing non-normal residuals

```{r}
# Simulate IV
x2 <- rnorm(n)

# Heavy-tailed distribution
errors_nonnormal <- rt(n, df = 2)
y_nonnormal <- 2 + 3*x2 + errors_nonnormal

# Fit model
model_nonnormal <- lm(y_nonnormal ~ x2)
```

Testing non-normal residuals



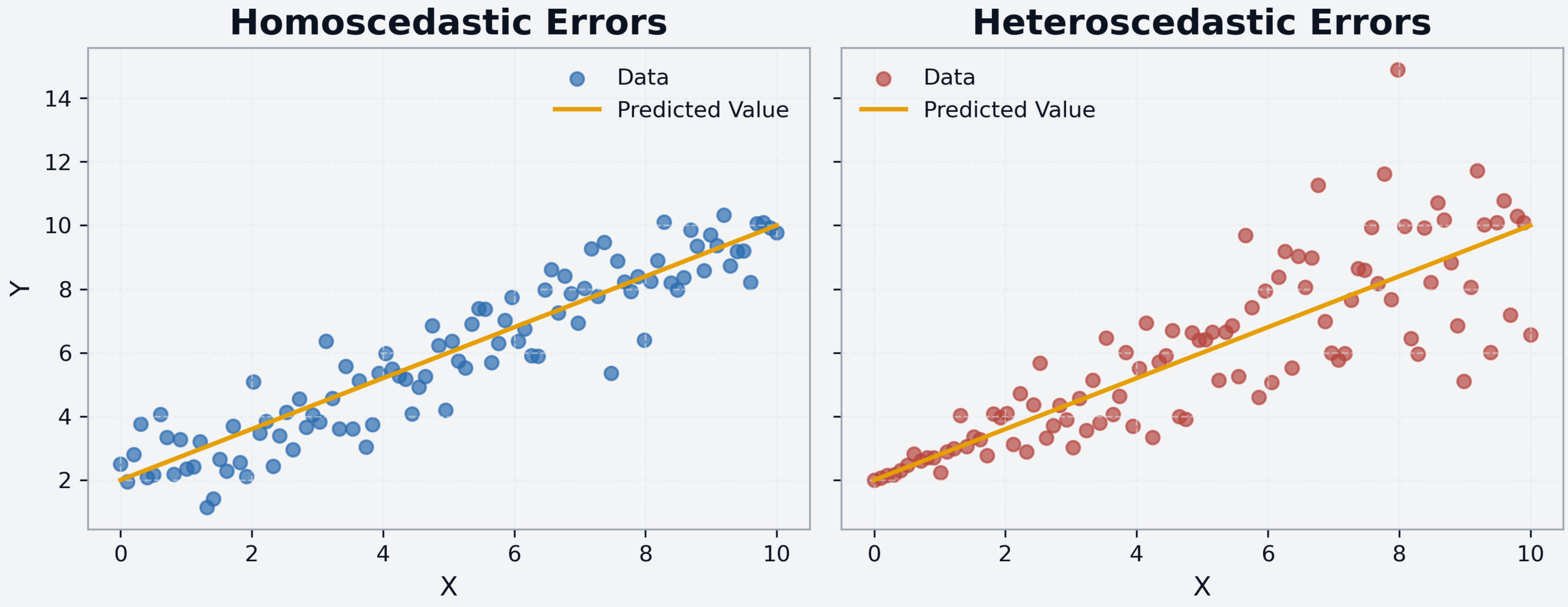
Solutions

- Solutions depends on the cause, but here are some possible solutions:
    - Collect more data
    - Transform non-linear variables (e.g. log(GDP))
    - Use a different type of regression (e.g. logistic, poisson. Not covered in this class)
    - heteroskedasticity solutions (see following section)
    - Use non-parametric hypothesis testing like the permutation test

Heteroskedasticity

Heteroskedasticity

- Variance in the error term is not constant



Heteroskedasticity

- Heteroskedasticity violates Gauss-Markov assumptions
- Coefficients are unbiased, but inefficient (i.e. there are estimators other than OLS that produce estimates with lower variance)
- However, standard errors are biased leading to invalid hypothesis testing and incorrect confidence intervals

Causes of heteroskedasticity

- Omitted variables. If the effect of a variable varies across the data, and it is absorbed into the error term, it can induce heteroskedasticity
- Incorrect functional form (i.e. using a linear model for a non-linear process)
- Outliers and a wide range of values in your data.

Diagnosing heteroskedasticity

- Plot residuals and fitted values

- White's test

```
```{r}
x2 <- rnorm(n)

# Heteroskedastic errors: variance increases with x
errors_hetero <- rnorm(n, sd = abs(x2) * 2)

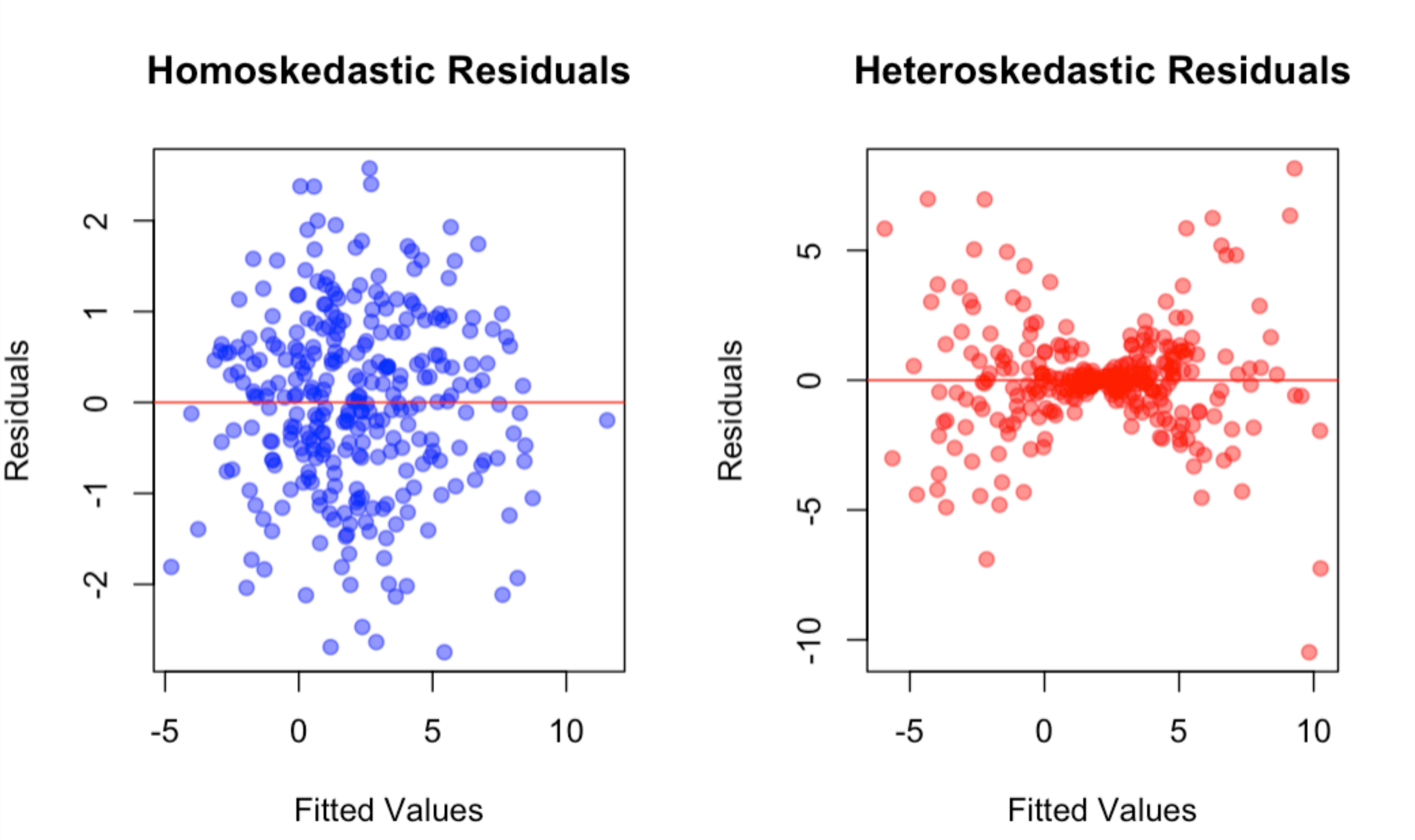
y_hetero <- 2 + 3*x2 + errors_hetero

model_hetero <- lm(y_hetero ~ x2)

# Plot residuals vs fitted values
plot(fitted(model_hetero), resid(model_hetero),
     main = "Residuals vs Fitted (Heteroskedastic Case)",
     xlab = "Fitted Values", ylab = "Residuals")
abline(h = 0, col = "red")

# White's test
white_hetero <- bptest(model_hetero, ~ fitted(model_hetero) + I(fitted(model_hetero)^2))
white_hetero
```
```

Diagnosing heteroskedasticity



Resolving heteroskedasticity

- Robust standard errors. Adjusts OLS standard errors to account for non-constant variance

```
```{r}

library(sandwich)
library(lmtest)

# White's robust SE
coeftest(model_nonnormal, vcov = vcovHC(model_nonnormal, type = "HC0"))
```
```

- Transform variables (e.g. log(GDP))

- Improve your model

- Weighted least squares (we won't get into this)

Multicollinearity

Multicollinearity

- Occurs when the predictors in a regression model are highly correlated
- Causes coefficients to be unstable, may vary widely with changes in data or model specification
- Standard errors become inflated, leading to higher p-values
- Can be difficult to isolate the effect of a correlated variable
- Ignorable if you don’t need to interpret the affected coefficient

Diagnosing multicollinearity

- Variance Inflation Factor > 5. VIF measures how much of a coefficient's variance is due to multicollinearity.

```{r}
library(car)
n <- 100
x1 <- rnorm(n)
x2 <- x1 + rnorm(n, sd = 0.1)  # highly correlated with x1
x3 <- rnorm(n)
y <- 2 + 3*x1 + 2*x2 + 1.5*x3 + rnorm(n)

# Fit linear model
model <- lm(y ~ x1 + x2 + x3)

vif_values <- vif(model)
vif_values
```

```
        x1         x2         x3 
115.214080 115.318824   1.033462 
```

- Unstable coefficients

- High R^2 with few statistically significant coefficients

Resolving multicollinearity

- Multicollinearity is primarily a sample size issue
- Do **not** drop predictors simply because they are highly correlated, no matter what google tells you.